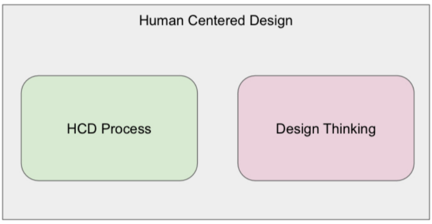
IUM

To make this more clear, any business can use Design Thinking to build a solution that is capable of making money.
For example, a company may use Design Thinking to create a video game or TV
show for kids.
Applying Human-Centered Design on top of this will ensure that the show actually serves the needs of the people watching it.



HCD is a mindset.
HCD Process e Design Thinking are design methods

Waterfall Development

Waterfall methodology is a linear project management approach, where stakeholder and customer **requirements are gathered at the beginning** of the project, and then a **sequential project plan** is created to accommodate those requirements.

The waterfall method is so named because **each phase of the project cascades** into the next, following steadily down like a waterfall.

It's a thorough, structured methodology and one that's been around for a long time, because it works. Some of the industries that regularly use the waterfall method include construction, IT and software development.

However, the term "waterfall" is usually used in a software context.

Agile

Agile is the ability to create and respond to change. It is a way of dealing with, and ultimately succeeding in, an uncertain and turbulent environment.

It's really about thinking through how you can understand what's going on in the environment that you're in today, identify what uncertainty you're facing, and figure out how you can adapt to that as you go along.

Agile software development is more than frameworks such as Scrum, Extreme Programming or Feature-Driven Development (FDD).

Agile software development is an umbrella term for a set of frameworks and practices based on the values and principles expressed in the Manifesto for Agile Software Development and the 12 Principles behind it.

Agile

One thing that separates Agile from other approaches to software development is the **focus on the people doing the work and how they work together.**

Solutions evolve through collaboration between self-organizing cross-functional teams utilizing the appropriate practices for their context.

There's a big focus in the Agile software development community on collaboration and the self-organizing team.

Agile is the best building method for HCD and Design thinking where continuous iterations are required

Scrum

Scrum is an agile framework for developing, delivering, and sustaining complex products, with an initial emphasis on **software** development, although it has been used in other fields including **research, sales, marketing and advanced technologies.**

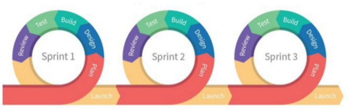
It is designed for **teams of ten or fewer members**, who break their work into **goals that can be completed within timeboxed iterations, called sprints**, no longer than one month and most commonly **two weeks**.

The Scrum Team track progress in 15-minute time-boxed daily meetings, called daily scrums. At the end of the sprint, the team holds sprint review, to demonstrate the work done, and sprint retrospective to improve continuously.

Scrum

A key principle of Scrum is the dual recognition that **customers will change their minds about what they want or need** (often called requirements volatility) and that there will be unpredictable challenges for which a predictive or planned approach is not suited.

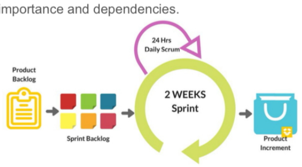
As such, **Scrum adopts an evidence-based empirical approach** – accepting that **the problem cannot be fully understood or defined up front**, and instead focusing on how to maximize the team's ability to deliver **quickly, to respond to emerging requirements**, and to adapt to evolving technologies and changes in market conditions



Scrum

There are three roles in the Scrum framework.

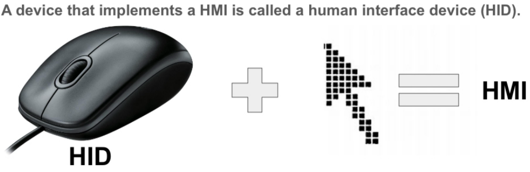
- **The product owner**, representing the product's stakeholders and the voice of the customer, is responsible for delivering good business results.The product owner defines the product in customer-centric terms (typically user stories), adds them to the Product Backlog, and prioritizes them based on importance and dependencies.
- **The development team**
- **The scrum master** is not a traditional team lead or project manager but acts as a buffer between the team and any distracting influences. The scrum master ensures that the scrum framework is followed.



Le interfacce utente

User interfaces are composed of one or more layers including a human-machine interface (HMI) interfaces machines with physical input hardware such a keyboards, mice, game pads and output hardware such as computer monitors, speakers, and printers.

A device that implements a HMI is called a human interface device (HID).



Possiamo quindi organizzare le interfacce in 6 categorie:

- Tactile UI
- Visual UI
- Auditory UI
- Olfactory UI
- Gustatory UI
- Equilibrial UI

1+ sensi => Compose user interface

Ex: Gui (vista, tatto); Mui (Gui + suono)

Tutti sensi => Qualia interface

Tipi di Cui:

Standard | Virtual | Augmented

All hardware innovations necessitated either overloading the use of data in an existing protocol or the creation of custom device drivers and the evangelization of a new protocol to developers. By contrast, all **HID-defined devices deliver self-describing packages that may contain any number of data types and formats.**

A single HID driver on a computer parses data and enables **dynamic association of data I/O with application functionality**, which has enabled rapid innovation and development, and prolific diversification of new human-interface devices.

The HID protocol has its limitations, but all modern mainstream operating systems will recognize standard USB HID devices, such as keyboards and mice, without needing a specialized driver. When installed, a message saying that "A 'HID-compliant device' has been recognized" generally appears on screen.

HID (software specification)

In the HID protocol, there are 2 entities: the **"host"** and the **"device"**.

The device is the entity that directly interacts with a human, such as a keyboard or mouse.

The host communicates with the device and receives input data from the device on actions performed by the human. Output data flows from the host to the device and then to the human.

Devices define their data packets and then present a "HID descriptor" to the host.

The HID descriptor is a hard coded array of bytes that describes the device's data packets.

This includes:

- how many packets the device supports,
- the size of the packets,
- the purpose of each byte and bit in the packet.

The device typically stores the HID descriptor in ROM and **does not need to intrinsically understand or parse the HID descriptor.** Some mouse and keyboard hardware in the market today is implemented using only an 8-bit CPU.

The host is expected to be a more complex entity than the device. **The host needs to retrieve the HID descriptor from the device and parse it before it can fully communicate with the device.**

Known buses that use HID are:

- Bluetooth HID – Used for mouse and keyboards that are connected via Bluetooth
- Serial HID – Used in Microsoft's Windows Media Center PC remote control receivers.
- ZigBee input device – ZigBee (RF4CE) supports HID devices through the ZigBee input device profile.
- HID over I²C – Used for embedded devices in Microsoft Windows 8[2]
- HOGP (HID over GATT) – Used for HID devices connected using Bluetooth low energy technology

HID (peripheral)

HID peripherals can be organized in two main categories:

- Input devices
- Output devices

Input devices are based on **sensors** (a device, module, or subsystem whose purpose is to detect events or changes in the physical world its environment and convert it in analog or digital electronic information)

Output devices are based on **actuators** (a device, module, or subsystem whose purpose is to convert analog or digital electronic signals in physical events aimed at changing the physical)

Keyboard layouts

A keyboard layout is any specific physical, visual or functional arrangement of the keys, legends, or key-meaning associations (respectively) of a computer keyboard, mobile phone, or other computer-controlled typographic keyboard.

Physical layout is the actual positioning of keys on a keyboard.

Visual layout the arrangement of the legends (labels, markings, engravings) that appear on those keys.

Functional layout is the arrangement of the key-meaning association or keyboard mapping, determined in software, of all the keys of a keyboard: this (rather than the legends) determines the actual response to a key press.

Barcode readers

A barcode reader (or barcode scanner) is an optical scanner **that can read printed barcodes**, decode the data contained in the barcode and send the data to a computer.

Like a flatbed scanner, it consists of a light source, a lens and a light sensor translating for optical impulses into electrical signals.

Additionally, nearly all barcode readers contain decoder circuitry that can analyze the barcode's image data provided by the sensor and sending the barcode's content to the scanner's output port.

Qrcode = bidim barcode

RFID

An RFID tag consists of a tiny radio transponder; a radio receiver and transmitter.

When triggered by an electromagnetic interrogation pulse from a nearby RFID reader device, the tag transmits digital data, usually an identifying inventory number, back to the reader. This number can be used to inventory goods.

There are two types:

- Passive tags are powered by energy from the RFID reader's interrogating radio waves.
- Active tags are powered by a battery and thus can be read at a greater range from the RFID reader; up to hundreds of meters.

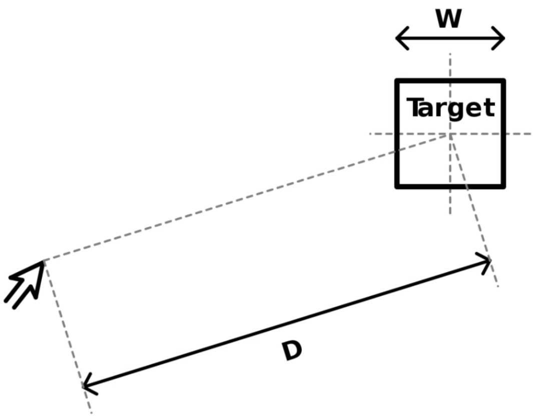
Unlike a barcode, the tag doesn't need to be within the line of sight of the reader, so it may be embedded in the tracked object.

Fitts's law

Movement Time:

- a = time to start/stop in seconds (empirically measured per device)
- b = inherent speed of the device (empirically measured per device)
- D is the distance from the starting point to the center of the target.
- W is the width of the target measured along the axis of motion.

$$\mathrm{MT} = a + b \cdot \mathrm{ID} = a + b \cdot \log_2\left(\frac{2D}{W}\right)$$

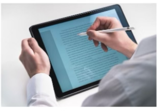


Pointing Devices classification

direct vs. indirect input

In case of a direct-input pointing device, **the on-screen pointer is at the same physical position as the pointing device** (e.g., finger on a touch screen, stylus on a tablet computer).

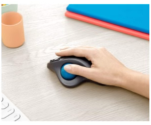
An indirect-input pointing device **is not at the same physical position as the pointer** but translates its movement onto the screen (e.g., computer mouse, joystick, stylus on a graphics tablet).



Pointing Devices classification

isotonic vs. elastic vs. isometric

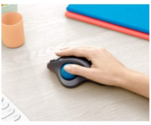
An isotonic pointing device **is movable and measures its displacement** (mouse, pen, human arm) whereas an isometric device **is fixed and measures the force which acts on it** (trackpoint, force-sensing touch screen). **An elastic device increases its force resistance with displacement** (joystick).



Pointing Devices classification

position control vs. rate control

A position-control input device (e.g., mouse, finger on touch screen) **directly changes the absolute or relative position of the on-screen pointer.** A rate-control input device (e.g., trackpoint, joystick) **changes the speed and direction of the movement of the on-screen pointer.**



An eye tracker is a device for measuring eye positions and eye movement.

Eye tracking methods

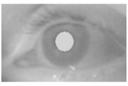
The most widely used current designs are **video-based eye-trackers.**

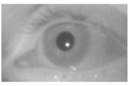
A camera focuses on one or both eyes and records eye movement. Most modern eye-trackers use the center of the pupil and infrared/near-infrared non-collimated light to create corneal reflections (CR).

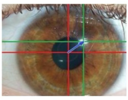
The vector between the pupil center and the corneal reflections can be used to compute the point of regard on surface or the gaze direction. A simple calibration procedure of the individual is usually needed before using the eye tracker.

Two general types of infrared/near-infrared (also known as active light) eye-tracking techniques are used: **bright-pupil (top) and dark-pupil (center).** Their difference is based on the location of the illumination source with respect to the optics.

Another, less used, method is known as **passive light (bottom)**. It uses visible light to illuminate







Bright-pupil tracking is more reliable but require more complex hardware setup

Eye-tracking vs. gaze-tracking

Eye-trackers necessarily measure the rotation of the eye with respect to some frame of reference. This is usually tied to the measuring system. Thus, if the measuring system is head-mounted, as with video-based system mounted to a helmet, then eye-in-head angles are measured.

To deduce the line of sight in world coordinates, the head must be kept in a constant position or its movements must be tracked as well. In these cases, head direction is added to eye-in-head direction to determine gaze direction.

If the measuring system is table-mounted, as with table-mounted camera ("remote") systems, then gaze angles are measured directly in world coordinates. Typically, in these situations head movements are prohibited.

Some results are available on human eye movements under natural conditions where head movements are allowed as well.

Data Glove

A wired glove (also called a "dataglove") is an input device for human–computer interaction worn like a glove.

Various sensor technologies are used to capture physical data such as bending of fingers.

Often a motion tracker, such as a magnetic tracking device or inertial tracking device, is attached to capture the global position/rotation data of the glove.

These movements are then interpreted by the software that accompanies the glove, so any one movement can mean any number of things.

Gestures can then be categorized into useful information, such as to recognize sign language or other symbolic functions.

Haptic devices allow users to touch, feel and manipulate three-dimensional objects in virtual environments and tele-operated systems.

Haptic devices are input-output devices, meaning that they track a user's physical manipulations (input) and provide realistic touch sensations coordinated with onscreen events (output)

Audio and sound acquisition

In physics, sound is a vibration that propagates as an acoustic wave, through a transmission medium such as a gas, liquid or solid.

In human physiology and psychology, sound is the reception of such waves and their perception by the brain.

Only acoustic waves that have frequencies lying between about 20 Hz and 20 kHz, the audio frequency range, elicit an auditory percept in humans.

In air at atmospheric pressure, these represent sound waves with wavelengths of 17 meters to 1.7 centimetres.

Sound waves above 20 kHz are known as ultrasound and are not audible to humans. Sound waves below 20 Hz are known as infrasound. Different animal species have varying hearing ranges.

A microphone array is any number of microphones operating in tandem.

Microphone Arrays

Typically, an array is made up of omnidirectional microphones, directional microphones, or a mix of omnidirectional and directional microphones distributed about the perimeter of a space

All microphones are linked to a computer that records and interprets the results into a coherent form.

Arrays may also be formed using numbers of very closely spaced microphones.

Given a fixed physical relationship in space between the different individual microphone transducer array elements, simultaneous DSP (digital signal processor) processing of the signals from each of the individual microphone array elements **can create one or more "virtual" microphones.**

Speech and Auditory Interfaces

Problem and limitations:

- Bandwidth is much lower than visual displays
- Ephemeral nature of speech (tone, etc.)
- Difficulty in parsing/searching → higher computational load

https://www.youtube.com/watch?v=IKZToY-V16w&ab_channel=EliyahuShatz

Speech and Auditory Interfaces

Succeed With:

Specialized vocabularies (like medical or legal)

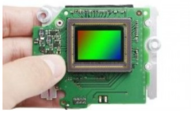
- Dictate reports, notes, letters
- Communication skills practice (virtual patient)
- Automatic retrieval/transcription of audio content (like radio, CC)
- Security/user ID

Image Sensors

The two main types of electronic image sensors are the charge-coupled device (CCD) and the active-pixel sensor (CMOS sensor).

Both CCD and CMOS sensors are based on metal–oxide–semiconductor (MOS) technology, with CCDs based on MOS capacitors and CMOS sensors based on MOSFET (MOS field-effect transistor) amplifiers.

Analog sensors for invisible radiation tend to involve vacuum tubes of various kinds.



3D scanning is the process of analyzing a real-world object or environment
to collect data on its shape and possibly its appearance (e.g. colour).

Many limitations in the kind of objects that can be digitised are still present. For
example, optical technology may encounter many difficulties with shiny, reflective or transparent objects.

Types of 3D scanners

Passive scanning: Passive 3D imaging solutions do not emit any kind of radiation themselves, but instead rely on detecting reflected ambient radiation.

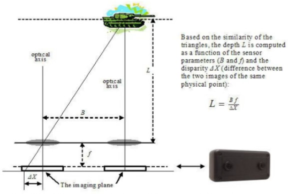
Most solutions of this type detect visible light because it is a readily available ambient radiation.

Other types of radiation, such as infrared could also be used. Passive methods can be very cheap, because in most cases they do not need particular hardware but simple digital cameras.

Stereoscopic cameras are the most common passive 3D scanning systems

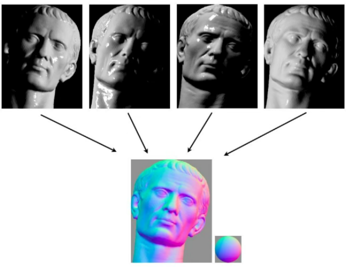
3D passive scanning

Stereoscopic systems usually employ two video cameras, slightly apart, looking at the same scene. By analysing the slight differences between the images seen by each camera, it is possible to determine the distance at each point in the images. This method is based on the same principles driving human stereoscopic vision.



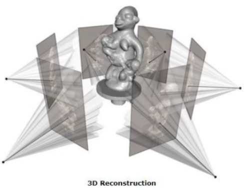
3D passive scanning

Photometric systems usually use a single camera, but take multiple images under varying lighting conditions. These techniques attempt to invert the image formation model in order to recover the surface orientation at each pixel.



3D passive scanning

Silhouette techniques use outlines created from a sequence of photographs around a three-dimensional object against a well contrasted background. These silhouettes are extruded and intersected to form the visual hull approximation of the object. With these approaches some concavities of an object (like the interior of a bowl) cannot be detected.

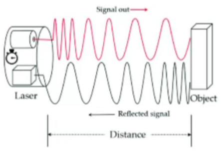


3D Active Scanning

The time-of-flight 3D laser scanner is an active scanner that uses laser light to probe the subject. At the heart of this type of scanner is a time-of-flight laser range finder.

The laser range finder finds the distance of a surface by timing the round-trip time of a pulse of light. A laser is used to emit a pulse of light and the amount of time before the reflected light is seen by a detector is measured. 3.3 picoseconds (approx.) is the time taken for light to travel 1 millimetre.

The laser range finder only detects the distance of one point in its direction of view. Thus, the scanner scans its entire field of view one point at a time by changing the range finder's direction

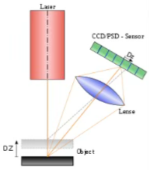


3D Active Scanning

Triangulation based 3D laser scanners are also active scanners that use laser light to probe the environment. With respect to time-of-flight 3D laser scanner the triangulation laser shines a laser on the subject and exploits a camera to look for the location of the laser dot.

Depending on how far away the laser strikes a surface, the laser dot appears at different places in the camera's field of view.

This technique is called triangulation because the laser dot, the camera and the laser emitter form a triangle.



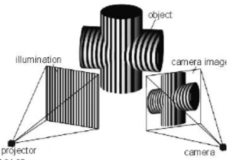
3D Active Scanning

Structured-light 3D scanners project a pattern of light on the subject and look at the deformation of the pattern on the subject. The pattern is projected onto the subject using either an LCD projector or other stable light source.

A camera, offset slightly from the pattern projector, looks at the shape of the pattern and calculates the distance of every point in the field of view.

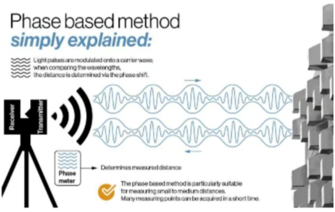
The advantage of structured-light 3D scanners is speed and precision. Instead of scanning one point at a time, structured light scanners scan multiple points or the entire field of view at once.

Scanning an entire field of view in a fraction of a second reduces or eliminates the problem of distortion from motion.



3D Active Scanning

Modulated light 3D scanners shine a continually changing light at the subject. Usually the light source simply cycles its amplitude in a sinusoidal pattern. A camera detects the reflected light and the amount the pattern is shifted by determines the distance the light travelled. Modulated light also allows the scanner to ignore light from sources other than a laser, so there is no interference.



Wii Remote and similar Motion Sensing devices

The Wii Remote, is the primary game controller for Nintendo's Wii home video game console.

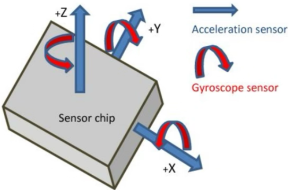
An essential capability of the Wii Remote is its motion sensing capability, which allows the user to interact with and manipulate items on screen via gesture recognition and pointing, using IMU and optical sensor technology.

The Wii Remote was eventually succeeded by the more advanced Joy-Con controllers of the Nintendo Switch.

IMU Inertial Measurement Unit

An inertial measurement unit (IMU) is an electronic device that measures and reports a body's **specific force, angular rate**, and sometimes the **orientation of the body**, using a combination **of accelerometers, gyroscopes, and sometimes magnetometers.**

9-Axis devices combine a **3-axis gyroscope, 3-axis accelerometer and 3-axis compass** (magnetometer) in the same chip together with an onboard Digital Motion Processor capable of processing the complex MotionFusion algorithms



A wearable computer is a computing device worn on the body. The interface and
the computing unit are merged together!
Wearables may be for general use, in which case they are just a particularly small
example of mobile computing. Alternatively they may be for specialized purposes
such as fitness trackers.

Devices carried in a pocket or bag – such as smartphones and before them pocket
calculators and PDAs, may or may not be regarded as 'worn'. Wearable computers have various technical issues common to other mobile computing, such as batteries, heat dissipation, software architectures,

wireless and personal area networks, and data management

Heart Rate Wearable Monitor

A heart rate monitor (HRM) is a personal monitoring device that allows one to measure/display heart rate in real time or record the heart rate for later study.

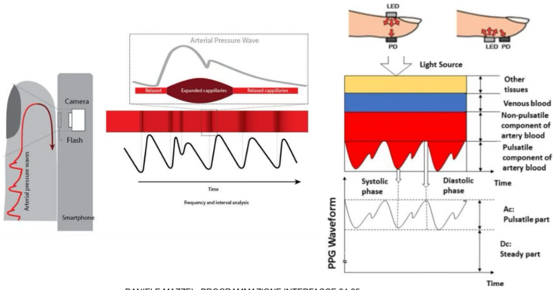
Consumer heart rate monitors are designed for everyday use and do not use wires to connect and are (typically) not suitable for medical applications

they commonly use electrical and/or optical measurement principles. Both types of signals can provide the same basic heart rate data, using fully automated algorithms to measure heart rate.

ECG (Electrocardiography) sensors measure the bio-potential generated by electrical signals that control the expansion and contraction of heart. VIDEO

PPG (Photoplethysmography) sensors use a light-based technology to measure the blood volume controlled by the heart's pumping action. some devices using this technology are able to measure aslo blood oxygen saturation (SpO2). more info

PPG (Photoplethysmography)



EEG Headset

Electroencephalography (EEG) is a monitoring method to record the electrical activity of the brain.

Wearable EEG headsets position noninvasive electrodes along the scalp. The clinical definition of EEG is the recording of brain activity over a period of time. EEG electrodes pick up on and record the electrical activity in your brain.

The collected signals are amplified and digitized then sent to a computer or mobile device for storage and data processing.

EEG Headset are used as pointing devices for dibale and impaired subjects

Interface structures

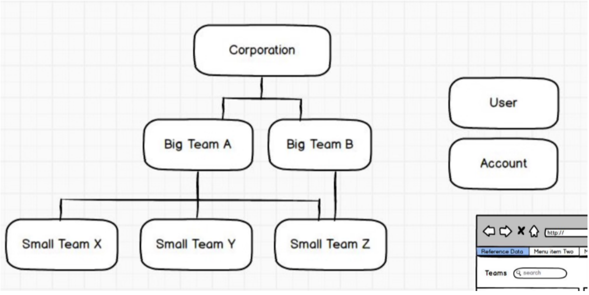
An organizational structure is how you define the **relationships between** pieces of **content**.

Successful structures allow users to predict where they will find information on the site.

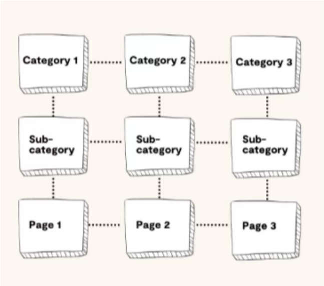
It’s important to take into account user expectations and implement consistent methods of organizing and displaying information so that users can extend their knowledge from familiar pages to unfamiliar ones.

The four main organizational structures are **Hierarchical, Sequential, Matrix and database model.**

source: https://www.usability.gov/how-to-and-tools/methods/organization-structures.html



An example of this type of structure is when a user is attempting to purchase something or are taking a course online. Sequential structures assume that there is some optimal ordering of content that is associated with greater effectiveness or success.



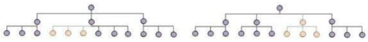
Database Model

The Database Model takes a bottom-up approach. The content within this structure leans heavily on the linkages created through the content's metadata. This type of model facilitates a more dynamic experience generally allowing for advanced filtering and search capabilities as well as providing links to related information in the system that has been properly tagged.

Creating Sustainable Structures

Site/GUI architecture has a long term impact on the site. It's important to put thought into the structure and ensure that it takes into account content updates in the future. Site managers should keep in mind the following when structuring a site:

Allow room for growth. Creating a site that can accommodate the addition of new content within a section (left image) as well as entire new sections (right image).

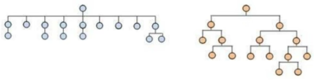


Creating Sustainable Structures

Avoid structures that are too shallow or too deep.

Striking a balance is never easy is an important goal of any architecture. Structures that are too shallow require massive menus.

Users rely on information architects to create logical groupings to facilitate movement throughout the site. In contrast, structures that are too deep bury information beneath too many layers. These structures burden the user to have to navigate through several levels to find the content that they desire.



Information Architecture

Information architecture (IA) focuses on organizing, structuring, and labeling content in an effective and sustainable way. The goal is to **help users find information** and complete tasks.

To do this, you need to understand how the pieces fit together to create the larger picture, how items relate to each other within the system.

Why a Well Thought Out IA Matters?

According to Peter Morville Site exit disclaimer, the **purpose of your IA is to help users understand where they are, what they've found, what's around, and what to expect.** As a result, your **IA informs the content strategy** through identifying word choice as well as informing user interface design and interaction design through playing a role in the wireframing and prototyping processes.

Information Architecture

To be successful, you need a diverse understanding of industry standards for creating, storing, accessing and presenting information. Lou Rosenfeld and Peter Morville in their book, Information Architecture for the World Wide Web, note that the main components of IA are:

- **Organization Schemes and Structures:** How you categorize and structure information
- **Labeling Systems:** How you represent information
- **Navigation Systems:** How users browse or move through information
- **Search Systems:** How users look for information

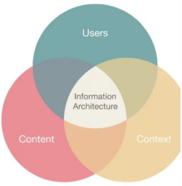
Information Architecture

In order to create these systems of information, you need to understand the interdependent nature of users, content, and context.

Rosenfeld and Morville referred to this as the "information ecology" and visualized it as a venn diagram.

Each circle refers to:

- **Context:** business goals, funding, politics, culture, technology, resources, constraints
- **Content:** content objectives, document and data types, volume, existing structure, governance and ownership
- **Users:** audience, tasks, needs, information-seeking experience



Exact Organization Schemes

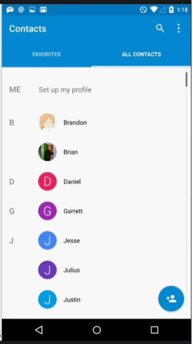
Exact organization schemes objectively **divide information into mutually exclusive sections**.

These systems comparatively are easy for information architects to create and categorize content within. However, they can be a challenge at times for users.

It requires that the user understands how what they are looking for fits within the model.

Examples of exact organizational structures include:

Exact Organization Schemes

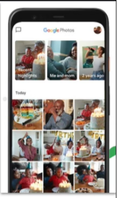


Alphabetical schemes make use of our 26-letter alphabet for organizing their contents.

For this type of scheme to be successful, it is important that the content labeling matches the words that users are looking for.

Sometimes, alphabetical schemes in the form of an A-Z index serve as secondary navigational components to supplement content findability that is otherwise organized.

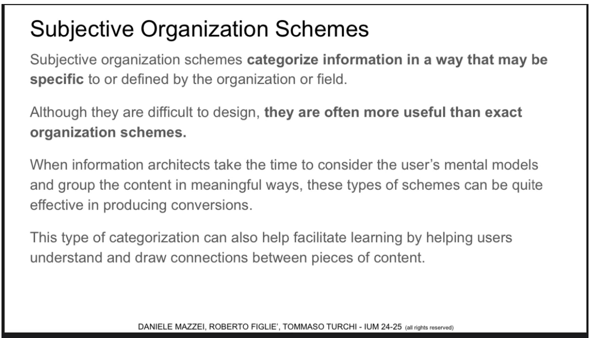
Exact Organization Schemes

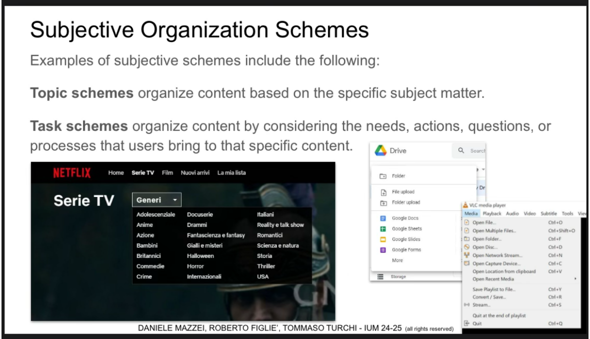


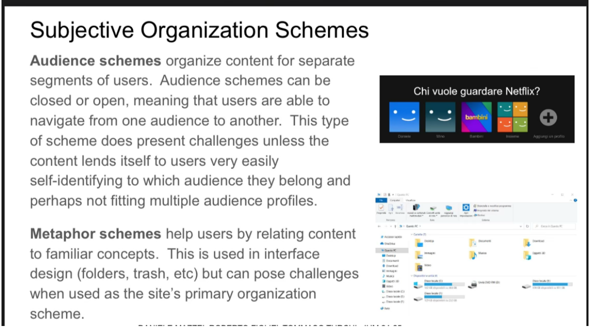
Chronological schemes organize content by date. For these schemes to be successful there must be agreement about when the subject of the content took place.

Geographical schemes organize content based on place. Unless there are border disputes, this type of scheme is fairly straightforward to design and use.

Often these types of schemes serve as a good supplemental way to navigate a site that that is otherwise organized. For example, you may choose to provide a map to display information or an A-Z index to get to topics grouped primarily by one of the following subjective schemes.







Navigation is about the user's orientation in the interface, and its main goal is to enable the user to find the information and functions he or she is looking for and, not only that, to encourage him or her in a direction that might be desirable to him or her. The basic principles for good navigation are findability and discoverability.

Findability means being able to get to the information you are looking for, or more precisely, how the user is able to localize the information he or she considers revelant.

Discoverability refers to the possibility that users have of discovering new information or new features that they were not aware of and were not specifically looking for.

With Findability the users knows or expects that a certain content or feature will be available.

With Discoverability the users don’t know that the content or functionality is there: they have to discover it!